

A Microscopy-Based Approach to Characterizing Particulate Contaminants in Snowfall on Suburban Long Island, New York

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Abstract:

Snowfall can act as an effective sampler of atmospheric particulates, capturing a wide range of contaminants that are later released into the environment during snowmelt. This study provides a preliminary microscopic characterization of particulate contaminants in suburban snowfall, establishing a methodological framework for future geochemical analysis of heavy metal deposition in winter precipitation.

Snow samples were collected from a suburban site on Long Island, specifically the rooftop of Gittleson Hall at Hofstra University in Hempstead, New York, during two snowfall events in 2026. Controlled collection procedures were used to minimize contamination, followed by melting, filtration, and laboratory preparation.

Collected particulates were analyzed using optical microscopy and scanning electron microscopy (SEM) to examine morphology, size distribution, and aggregation patterns. Observations revealed a heterogeneous mixture of contaminants, including fine particles, fibrous materials, and irregularly shaped aggregates. SEM imaging identified clustered and porous structures, suggesting the presence of complex, multi-source atmospheric particulates likely influenced by urban and suburban activities.

Although elemental composition was not determined in this phase of the study, the observed diversity of particles indicates multiple potential sources, including anthropogenic emissions (such as road salt) and natural debris. These findings support the role of snowfall as a temporary reservoir for atmospheric contaminants and highlight its importance in transporting particulate matter into terrestrial and aquatic systems during melt events.

Introduction:

Snowfall is widely recognized as an efficient collector of atmospheric particulate matter, capturing aerosols, dust, and anthropogenic pollutants during deposition and subsequently releasing them during melt events. This process plays an important role in the redistribution of contaminants from the atmosphere to terrestrial and aquatic systems, while surface-derived particles may also be reintroduced into the atmosphere through resuspension processes such as wind erosion and urban disturbance.

Numerous studies worldwide have demonstrated that snow can accumulate a variety of pollutants, including particulate matter, black carbon, road salts, and heavy metals such as lead (Pb), zinc (Zn), and copper (Cu), particularly in urban and industrial environments. Bakatsias et al. (2025) examined contaminant mobility in groundwater at two Superfund sites in New York State: Dzus Fasteners Co. in

West Islip, Long Island, and American Thermostat Co. in South Cairo (Town of Catskill), Greene County. Urban regions such as New York City are especially susceptible to elevated particulate and metal contamination due to high traffic density, dense transportation networks, and legacy urban pollution. Previous studies have documented elevated lead concentrations in urban soils across major U.S. cities, including New York City, where historical use of leaded gasoline, paint, and industrial emissions has contributed to persistent environmental contamination (Mielke et al., 2011; Laidlaw et al., 2012). These contaminants can be resuspended into the atmosphere and subsequently incorporated into precipitation, including snowfall, further contributing to environmental redistribution.

Previous research in Europe, Asia, and North America has shown that snowpack composition reflects local emission sources, including vehicular traffic, industrial activity, and residential heating (Westerlund et al., 2003). In the United States, studies have documented the role of snowfall and snowmelt in transporting atmospheric contaminants and influencing runoff quality, particularly in urbanized regions where road salt and particulate pollution are prevalent (Kaushal et al., 2005). Snow has also been shown to act as a temporary reservoir for airborne particulates, concentrating contaminants that may later impact soil and water systems.

Despite extensive research on snow chemistry and atmospheric deposition at broader regional and global scales, there is limited site-specific characterization of particulate contaminants in snowfall across suburban environments such as Long Island, New York. In particular, few studies have focused on the microscopic morphology and physical characteristics of particles captured in snowfall, which can provide important insight into contaminant sources, transport mechanisms, and aggregation processes.

This study presents a preliminary investigation of particulate contaminants collected from snowfall at a suburban site in Hempstead, New York (Hofstra University campus). Using optical microscopy and scanning electron microscopy (SEM), this work focuses on characterizing particle morphology, size distribution, and aggregation patterns. By establishing a microscopy-based framework, this study aims to provide foundational insight into snowfall-mediated particulate deposition and to support future research incorporating elemental analysis and multi-site comparisons across urban and suburban environments.

Snowfall and snowmelt in New York City and Long Island are capable of accumulating particulate contaminants, including potentially heavy metals and urban pollutants that leach from urban pollution, which poses significant risks to water quality and public health. Snow in a suburban site on Long Island acts like an “effective collector”) absorbing heavy metals such as copper, zinc, and lead along with other urban pollutants before releasing a concentrated aggregation of contaminants during melting. This release of contaminants can harm aquatic ecosystems and pose public-health risks as heavy-metal-laden runoff enters streams, rivers, and coastal waters. This study investigates the concentration of these contaminants through the collection of snow samples, melting them, and analyzing their morphology and distribution using optical microscopy and SEM, with future work aimed at determining elemental composition.

Methodology

Snow samples were collected from the top of *Gittleson Hall* at Hofstra University from 2 separate snowfall events. The first set of samples was collected on January 28, 2026, and the second set of samples was collected on February 25, 2026. In order to reduce the chance for sample contamination from sources like soil, debris, and surface water run-off, samples were collected from the middle layer of the snowpack to avoid contamination from both surface contact and post-depositional atmospheric exposure. In addition, the top 1 inch of the snowfall was avoided in order to reduce sample contamination from airborne sources after the snowfall. This method allowed for the analysis of the middle part of the snowfall samples, which was most representative of any pollution that was accumulated during the snowfall event. Clean collection equipment was used during the collection procedure in order to maintain consistency among all sample collection sites. Once collected, the snow samples were transported for melting, filtration, and to analyze the morphology, size distribution, and spatial clustering of particulate contaminants



Figure 1: Snow sampling procedure on the rooftop of Gittleson Hall, Hofstra University (Hempstead, NY) during the January 28, 2026 snowfall event. Samples were collected from the undersnowpack to minimize contamination from surface runoff and post-depositional atmospheric inputs.

During the initial phase of the project, a preliminary filtration system was constructed by modifying a large water container and integrating it into a laboratory sink setup. This system was used to process water samples and refine collection and filtration techniques. Building on this approach, the study was designed to collect snowfall and snowmelt samples from a typical suburban site near Hempstead NY how density and land-use characteristics influence pollutant deposition in winter precipitation.

Following method development, collected samples were prepared and analyzed using optical microscopy and scanning electron microscopy (SEM) to characterize particulate morphology and distribution. Although the scope of analysis was constrained to microscopic characterization, these observations provided valuable insight into the physical properties and diversity of particulate matter present in suburban snowfall. This iterative approach allowed for refinement of sampling and analytical methods while establishing a foundation for continued investigation. Future work will expand on these methods to further evaluate spatial variability and contaminant sources across urban and suburban environments.



Figure 2: Schematic diagram of the filtration setup used for preliminary sample processing, illustrating the modified container and filter system used to collect and concentrate particulate matter from melted snow samples.

For scanning electron microscopy (SEM) preparation, a small volume of clean water was carefully applied to the filter surface to mobilize and direct the sediment toward the middle region of the filter for the purposes of the SEM. Although complete consolidation of all particulates was not achievable due to some particles sticking in the filter, the majority of the sediment was successfully concentrated into a more uniform area to facilitate efficient SEM imaging and analysis. The filter was then allowed to dry completely. Once dried, the central portion containing the concentrated particulates was trimmed to an appropriate size and mounted onto an SEM specimen stub using conductive adhesive to reduce charging effects. The prepared specimen was introduced into the SEM chamber, vacuum conditions were established, and imaging parameters were optimized through the instrument's control software prior to conducting high-resolution morphological characterization of the collected contaminants.

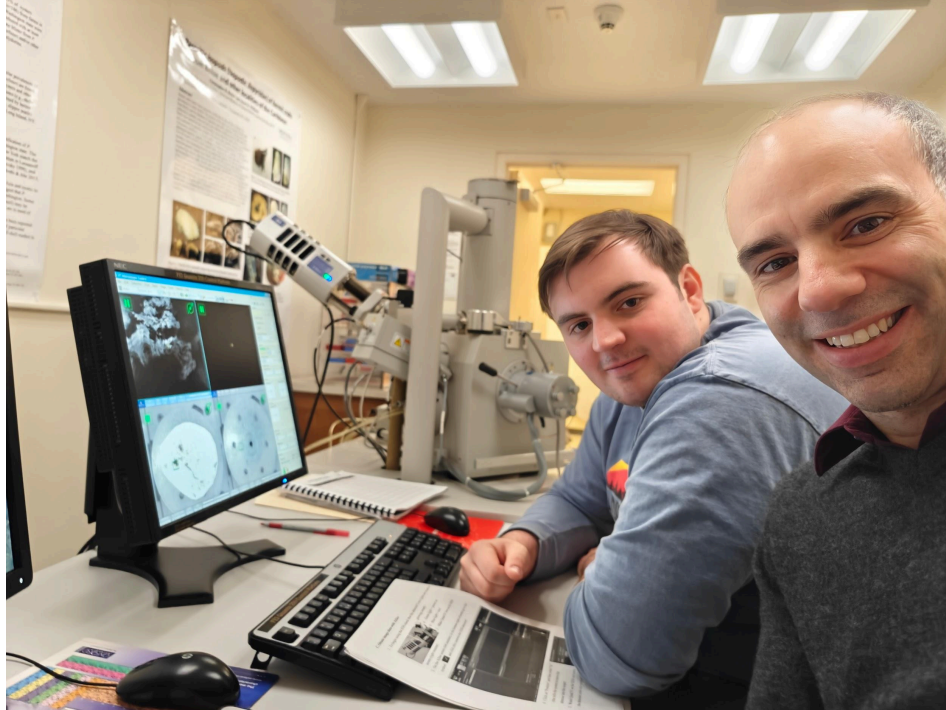


Figure 3: Dominick Colucci (left) and Dr. Antonios Marsellos conducting preliminary analysis of snow-derived particulate contaminants using a scanning electron microscope (SEM) at Hofstra University.

Results

Sufficient particulate material was collected for analysis. Using the setup of the filter experiment, we were able to garner sufficient results from the experiment. From the in lab procedure, we detected dark aggregated particles, fibrous material, and granular debris through the filter experiment. The particles varied in size from fine to coarse aggregates. Based on the results through the filter, there was an indication of multiple contamination sources, including atmospheric deposition, urban dust, and organic debris.

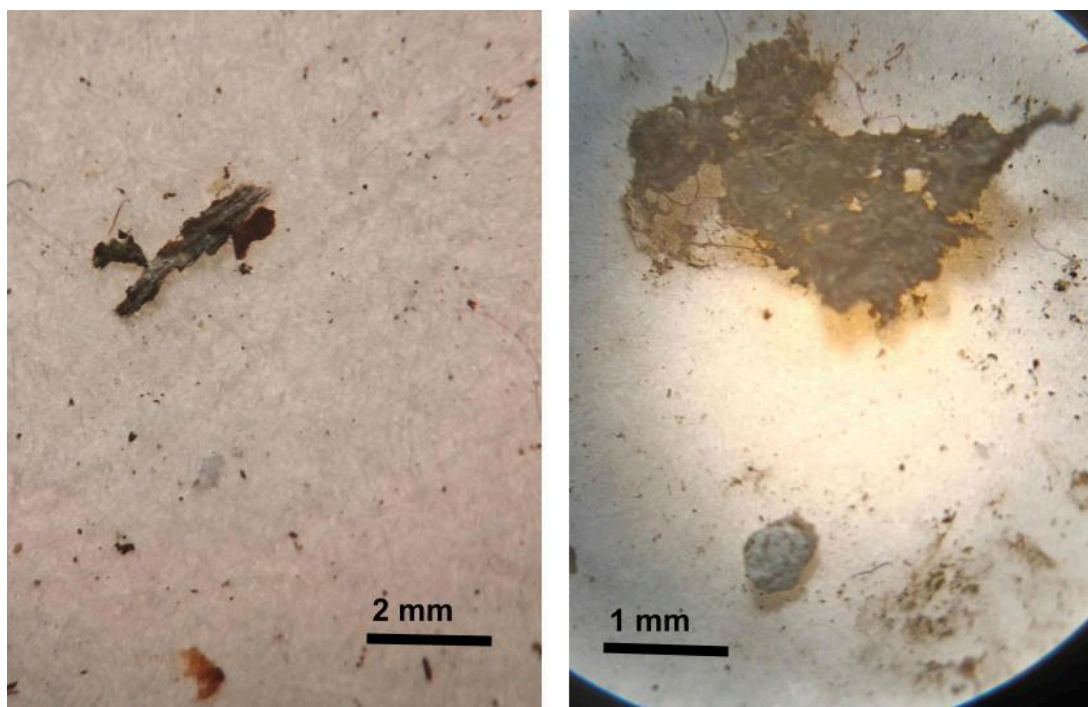


Figure 4: Optical microscopy images of particulate contaminants extracted from snowfall samples, showing heterogeneous material including dark aggregated particles, fibrous debris, and granular components. Scale bars indicate millimeter-scale variation in particle size.

Under the microscope, it was observed that the snow samples contained different kinds of particulate contaminants. The more prominent impurities observed were dark aggregated particles, which might be organic or textile in nature. The size of these impurities ranges from smaller impurities of sizes less than 100 μm , to larger impurities larger than 500 μm . The presence of granular impurities, which might be mineral dust, was also observed. The particles consistent with road-derived materials (e.g., salt or asphalt) was also observed” like road-derived material, which might be due to anthropogenic sources

Despite some minor challenges during filter preparation, sufficient particulate material was collected for SEM analysis. SEM imaging revealed irregular, non-spherical particles forming porous, clustered aggregates with diffuse and filamentous surface textures (Figure 5). The particles varied in size from fine to coarse structures, with no well-defined crystalline features typical of primary mineral grains. The observed morphology indicates a mixture of particle types and suggests the presence of aggregated material formed through secondary processes. These characteristics are consistent with particles associated with urban atmospheric contamination. However, further SEM imaging at higher magnifications and energy-dispersive spectroscopy (EDS) analysis will be necessary to more accurately determine the composition and origin of the collected material.

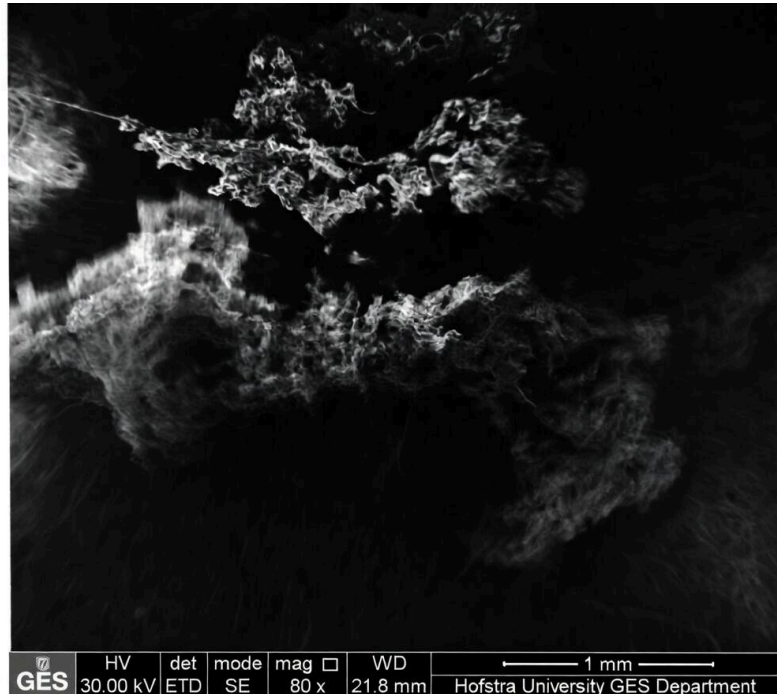


Figure 5: SEM Result of the same particle of figure 4 (right). SEM image shows aggregated particulate matter from snowfall samples, revealing irregular, non-spherical morphology with porous, clustered structures and filamentous textures.

Discussion

Figure 4 shows the filter after melting the snow which captures the dark aggregated particles, fibrous material, and granular debris we consistently observed in our samples. This visual indicates that suburban snowfall in Hempstead carries a mix of particulate matter collected during the snowfall event. The collected material serves as a reflection of multiple contamination sources, including atmospheric deposition and urban dust, as noted in our results. By documenting these particles at the filtration stage, the figure sets up the microscopic analysis that follows and reinforces snow's role as a temporary reservoir for urban contaminants.

In the optical microscopy image, a mixed composition of particulate material is visible, which was collected from the snowfall sample. The particulate matter includes dark-collected aggregated material, fibers, and granular material. The particulate matter is of different shapes and sizes, as some of the material is finer in texture and other material is present in clumps of bigger sizes. The presence of dark particulate matter might be due to contamination of the atmosphere due to urban areas, and the fibers might be due to organic or textile material contamination. The granular material might be due to dust or salt present on the road due to snowfall. The presence of a variety of particulate material in a single image

indicates that the snow might be collecting contaminants from various natural and human-related sources. The image also supports the idea of snowfall acting as a reservoir for particulate matter pollution.

The SEM image shows irregular, non-spherical particles forming porous, clustered aggregates with diffuse and filamentous surface textures, indicating a heterogeneous structure. These features suggest the particles are agglomerates formed through secondary processes such as atmospheric interactions rather than primary crystalline mineral grains. The variation in particle size and lack of defined crystal faces further support a mixed origin, consistent with urban atmospheric particulate matter. The morphology also implies a high surface area, which may enhance the interaction of particles with surrounding environmental compounds. Future research will include higher magnification SEM imaging and energy-dispersive spectroscopy (EDS) to better characterize particle structure and determine elemental composition. While this study focuses on a single suburban site, the methodology can be extended to broader spatial analyses across New York City and Long Island.

Conclusion

This study represents a pilot investigation of snowfall-mediated particulate deposition at a suburban Long Island site, providing a foundation for future multi-site and geochemical analyses. Our procedures clearly indicate that suburban snowfall collected on the Gittelsohn Hall rooftop at Hofstra University serves as a temporary reservoir for mixed urban particulates, with our optical and SEM analyses revealing dark aggregated particles, fibrous material, granular debris, and irregular clustered structures that point to diverse atmospheric contamination sources. These results suggest that snowfall can carry particles in suburban areas and move them when the snow melts. This shows why it is important to keep studying how these particles may affect nearby soil and water. Future work will involve higher magnification SEM imaging to further resolve particle morphology, along with energy-dispersive spectroscopy (EDS) to determine elemental composition. These analyses will provide a more detailed understanding of particle structure and composition.

Credit Authorship Contribution Statement

Colucci, D: snow sample collection, set up laboratory procedures, performed laboratory procedures, microscope procedure, SEM scanning, writing-Abstract, Introduction, Methods, Results, Discussion, Conclusion, editing

Schwartz, J: set up laboratory procedures, performed laboratory procedures, writing- Abstract, Introduction, Methods, Results, Discussion, Conclusion, editing

Singh, I: writing-Abstract, Introduction, Methods, Results, Discussion, Conclusion, editing

Marsellos, A.E.: supervision, research mentorship, sampling guidance.

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References

- Bakatsias, D.E., Kaur, J., & Meliker, J.R. (2025). Impact of rainfall, snowpack and storms on contaminant levels on groundwater beneath two U.S. Superfund sites in New York. *Journal of Environmental Exposure Assessment*, 4(39), <https://doi.org/10.20517/jeea.2025.50>
- Engelhard, C., De Toffol, S., Lek, I., Rauch, W., & Dallinger, R. (2007). Environmental impacts of urban snow management—The alpine case study of Innsbruck. *Science of the Total Environment*, 382(2–3), 286–294. <https://doi.org/10.1016/j.scitotenv.2007.04.008>
- Kaushal, S. S., Groffman, P. M., Likens, G. E., Belt, K. T., Stack, W. P., Kelly, V. R., ... Fisher, G. T. (2005). Increased salinization of fresh water in the northeastern United States. *Proceedings of the National Academy of Sciences*, 102(38), 13517–13520.
- Laidlaw, M. A. S., Zahran, S., Mielke, H. W., Taylor, M. P., & Filippelli, G. M. (2012). Re-suspension of lead contaminated urban soil as a dominant source of atmospheric lead in Birmingham, Chicago, Detroit and Pittsburgh, USA. *Atmospheric Environment*, 49, 302–310. <https://doi.org/10.1016/j.atmosenv.2011.10.030>
- Mielke, H. W., Gonzales, C. R., Powell, E. T., Laidlaw, M. A. S., Berry, K. J., Mielke, P. W., & Egendorf, S. P. (2019). The concurrent decline of soil lead and children's blood lead in New Orleans. *Proceedings of the National Academy of Sciences*, 116(44), 22058–22064.
- Westerlund, C., & Viklander, M. (2011). Pollutant release from a disturbed urban snowpack in northern Sweden. *Water Quality Research Journal of Canada*, 46(2), 98–109. <https://doi.org/10.2166/wqrjc.2011.025>
- Zhang, K., Chai, F., Zheng, Z., Yang, Q., Li, J., Wang, J., & Zhang, Y. (2014). Characteristics of atmospheric particles and heavy metals in winter in Chang-Zhu-Tan city clusters, China. *Journal of Environmental Sciences*, 26(1), 147–153.